In [22]:
```
import numpy as np
import pandas as pd
import os
for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

In [23]:
```
import seaborn as sns
import matplotlib.pyplot as plt
import scipy.stats as st
%matplotlib inline

sns.set(style="whitegrid")
```

In [24]:
```
import warnings
warnings.filterwarnings('ignore')
```

In [25]:
```
df = pd.read_csv(r'/Users/abhiramireddygali/Downloads/april(2025)/2r
```

In [26]:
```
print('The shape of the dataset : ',df.shape)
```

```
The shape of the dataset :  (303, 14)
```

In [27]:
```
df.head()
```

Out[27]:

	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slo
0	63	1	3	145	233	1	0	150	0	2.3	
1	37	1	2	130	250	0	1	187	0	3.5	
2	41	0	1	130	204	0	0	172	0	1.4	
3	56	1	1	120	236	0	1	178	0	0.8	
4	57	0	0	120	354	0	1	163	1	0.6	

In [28]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 303 entries, 0 to 302
Data columns (total 14 columns):
 #   Column    Non-Null Count  Dtype  
---  ------    --------------  -----  
 0   age       303 non-null    int64  
 1   sex       303 non-null    int64  
 2   cp        303 non-null    int64  
 3   trestbps  303 non-null    int64  
 4   chol      303 non-null    int64  
 5   fbs       303 non-null    int64  
 6   restecg   303 non-null    int64  
 7   thalach   303 non-null    int64  
 8   exang     303 non-null    int64  
 9   oldpeak   303 non-null    float64
 10  slope     303 non-null    int64  
 11  ca        303 non-null    int64  
 12  thal      303 non-null    int64  
 13  target    303 non-null    int64  
dtypes: float64(1), int64(13)
memory usage: 33.3 KB
```

In [29]:
```
df.dtypes
```

Out[29]:
```
age           int64
sex           int64
cp            int64
trestbps      int64
chol          int64
fbs           int64
restecg       int64
thalach       int64
exang         int64
oldpeak     float64
slope         int64
ca            int64
thal          int64
target        int64
dtype: object
```

In [30]:
```
df.describe()
```

Out[30]:

	age	sex	cp	trestbps	chol	
count	303.000000	303.000000	303.000000	303.000000	303.000000	303.(
mean	54.366337	0.683168	0.966997	131.623762	246.264026	0.
std	9.082101	0.466011	1.032052	17.538143	51.830751	0.:
min	29.000000	0.000000	0.000000	94.000000	126.000000	0.(
25%	47.500000	0.000000	0.000000	120.000000	211.000000	0.(
50%	55.000000	1.000000	1.000000	130.000000	240.000000	0.(
75%	61.000000	1.000000	2.000000	140.000000	274.500000	0.(
max	77.000000	1.000000	3.000000	200.000000	564.000000	1.(

In [31]:

```
df.columns
```

Out[31]:

```
Index(['age', 'sex', 'cp', 'trestbps', 'chol', 'fbs', 'restecg', '
thalach',
       'exang', 'oldpeak', 'slope', 'ca', 'thal', 'target'],
      dtype='object')
```

In [32]:

```
df['target'].nunique()
```

Out[32]:

```
2
```

In [33]:

```
df['target'].unique()
```

Out[33]:

```
array([1, 0])
```

In [34]:

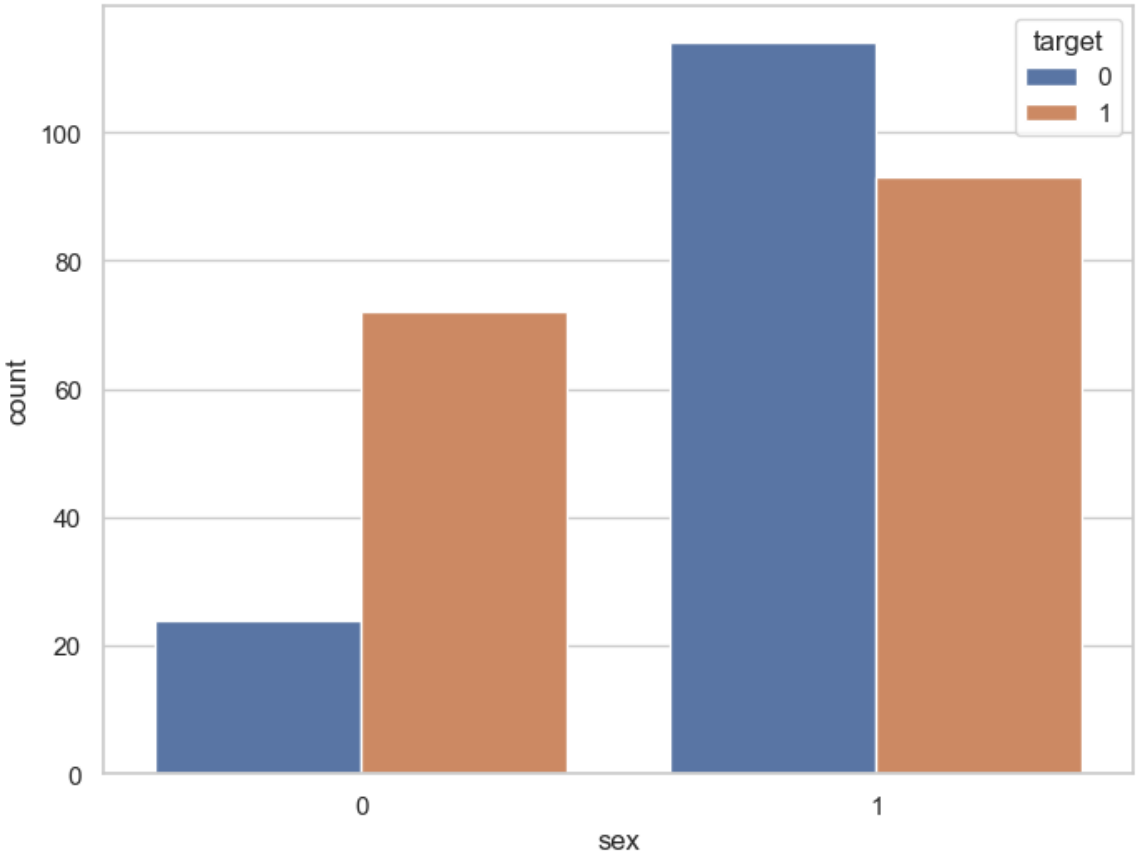
```
df['target'].value_counts()
```

Out[34]:

```
target
1    165
0    138
Name: count, dtype: int64
```

In [35]:

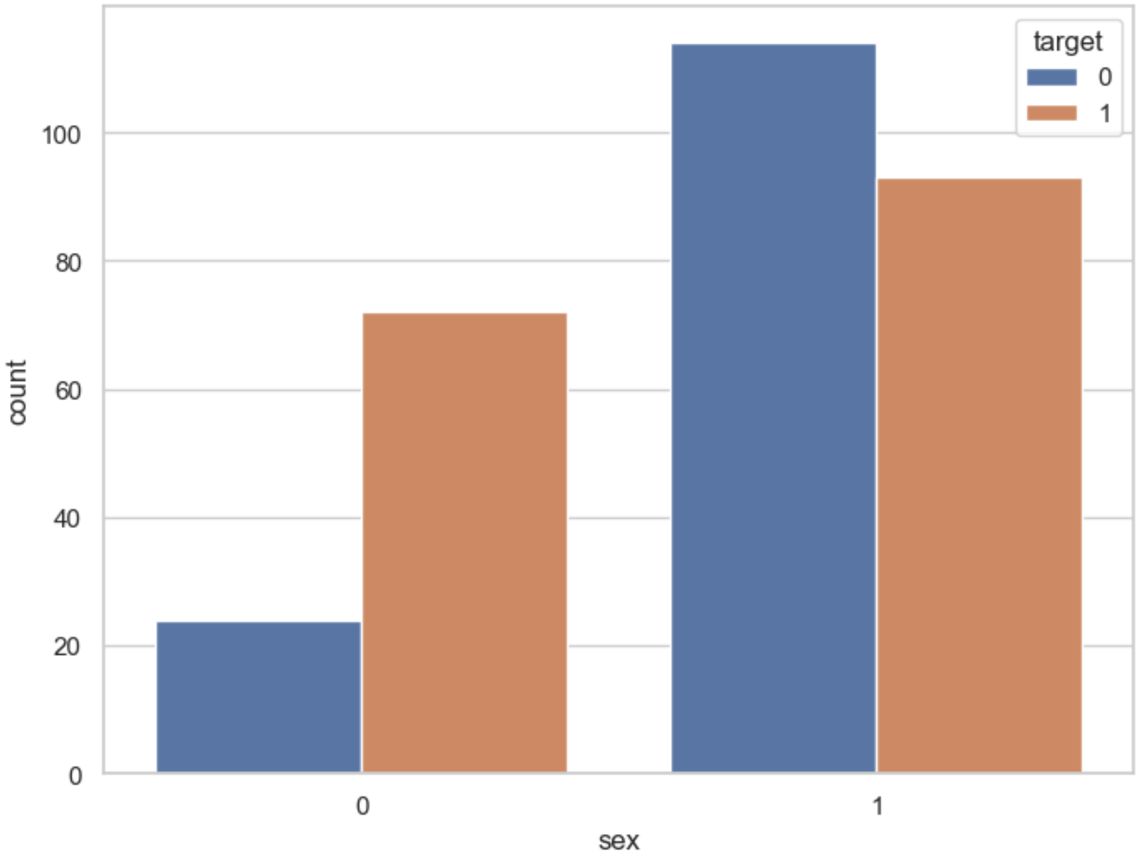
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="sex", hue="target", data=df)
plt.show()
```



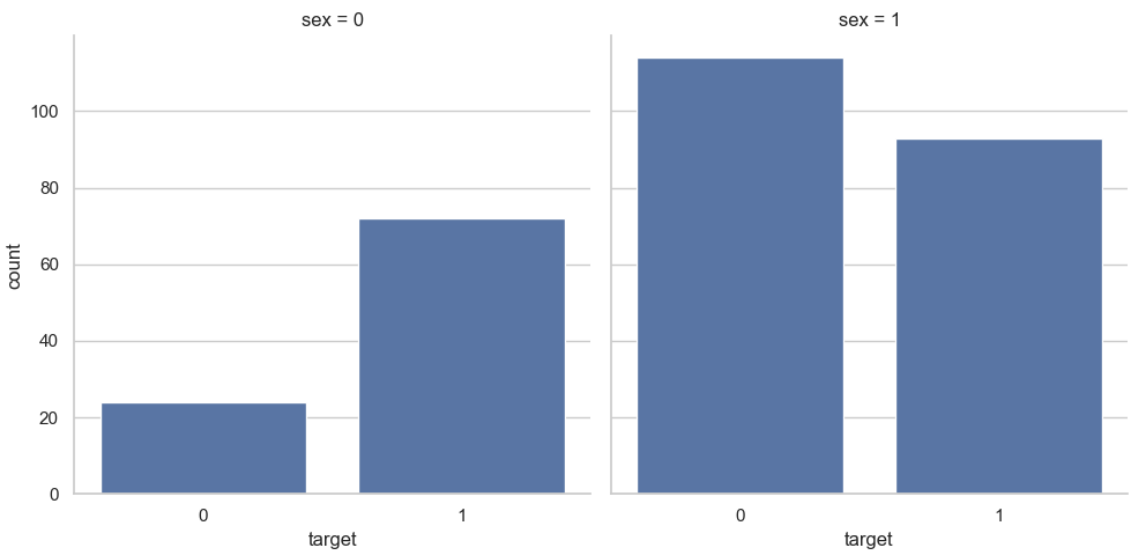
In [36]:
```
df.groupby('sex')['target'].value_counts()
```

Out[36]:
```
sex  target
0    1          72
     0          24
1    0         114
     1          93
Name: count, dtype: int64
```

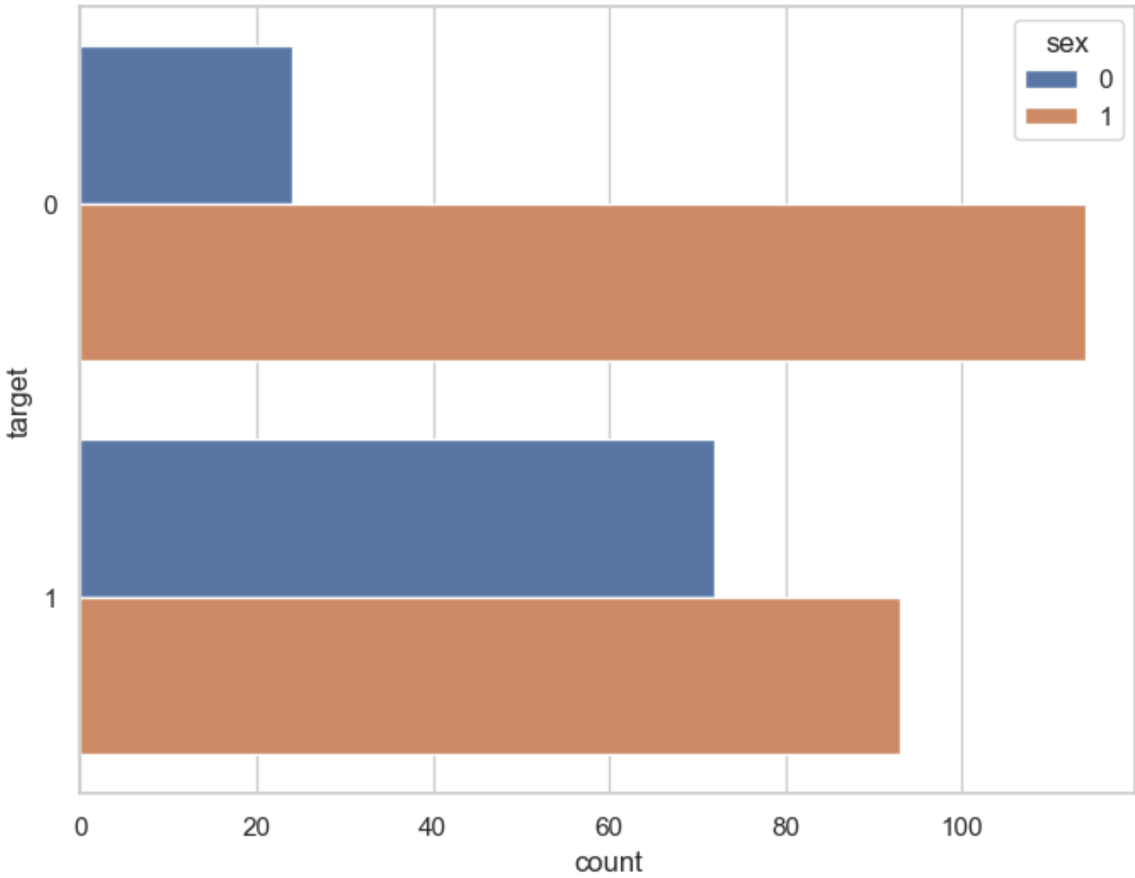
In [40]:
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="sex", hue="target", data=df)
plt.show()
```



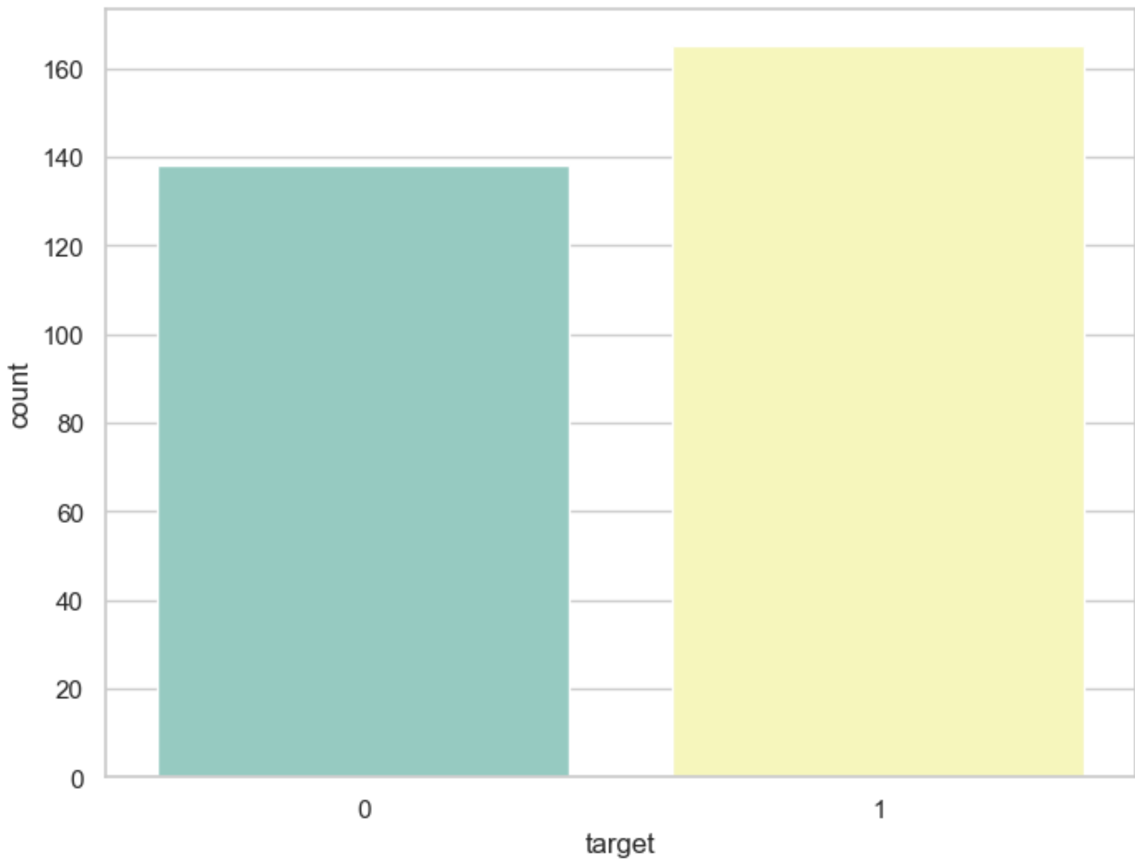
In [41]:
```
ax = sns.catplot(x='target',col ='sex',data =df,kind = 'count',heig
```



In [43]:
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(y="target", hue="sex", data=df)
plt.show()
```

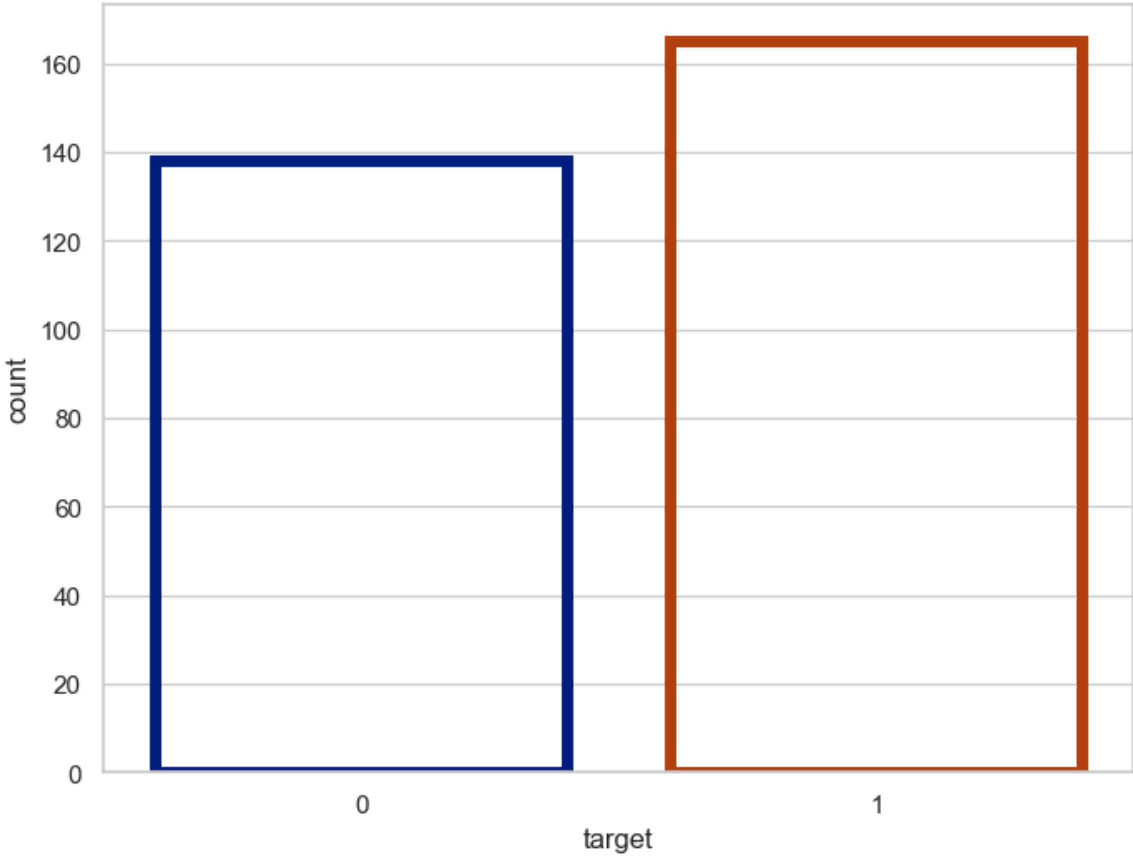


In [47]:
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(x="target", data=df, palette="Set3")
plt.show()
```

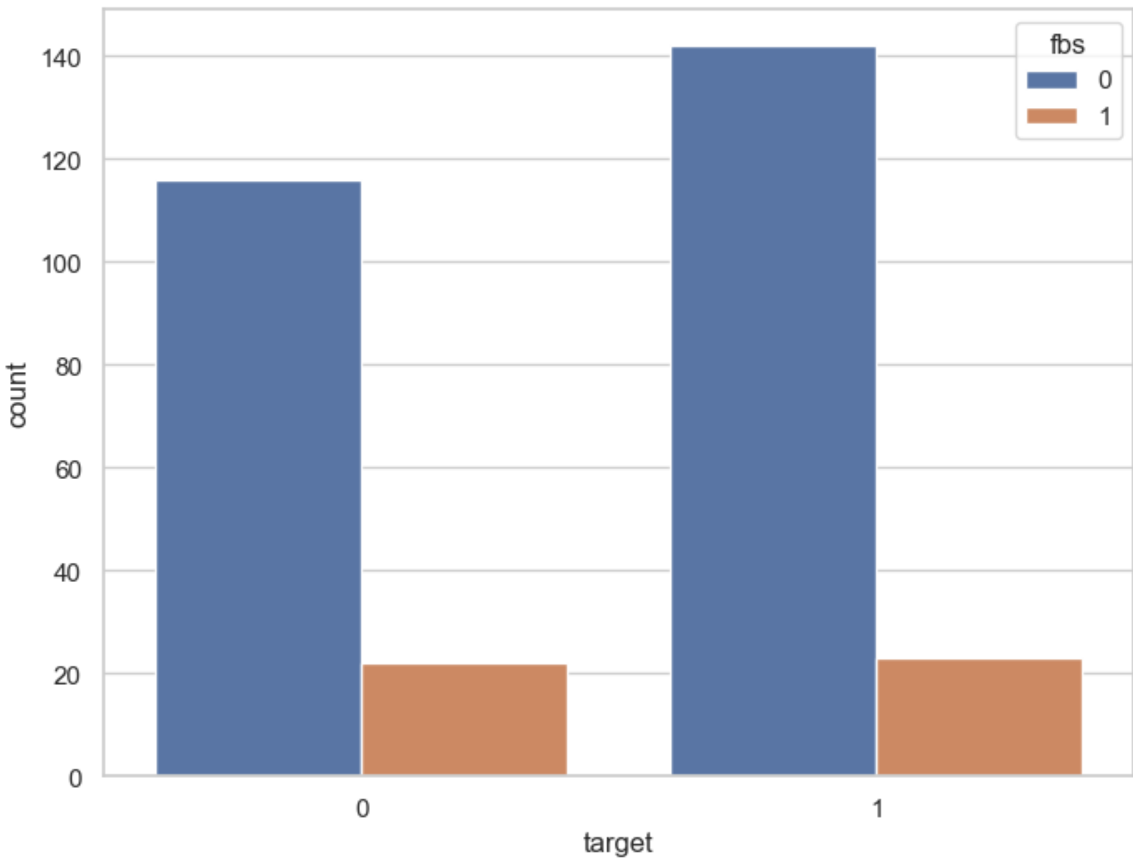


In [48]:
```
f, ax = plt.subplots(figsize=(8, 6))
```

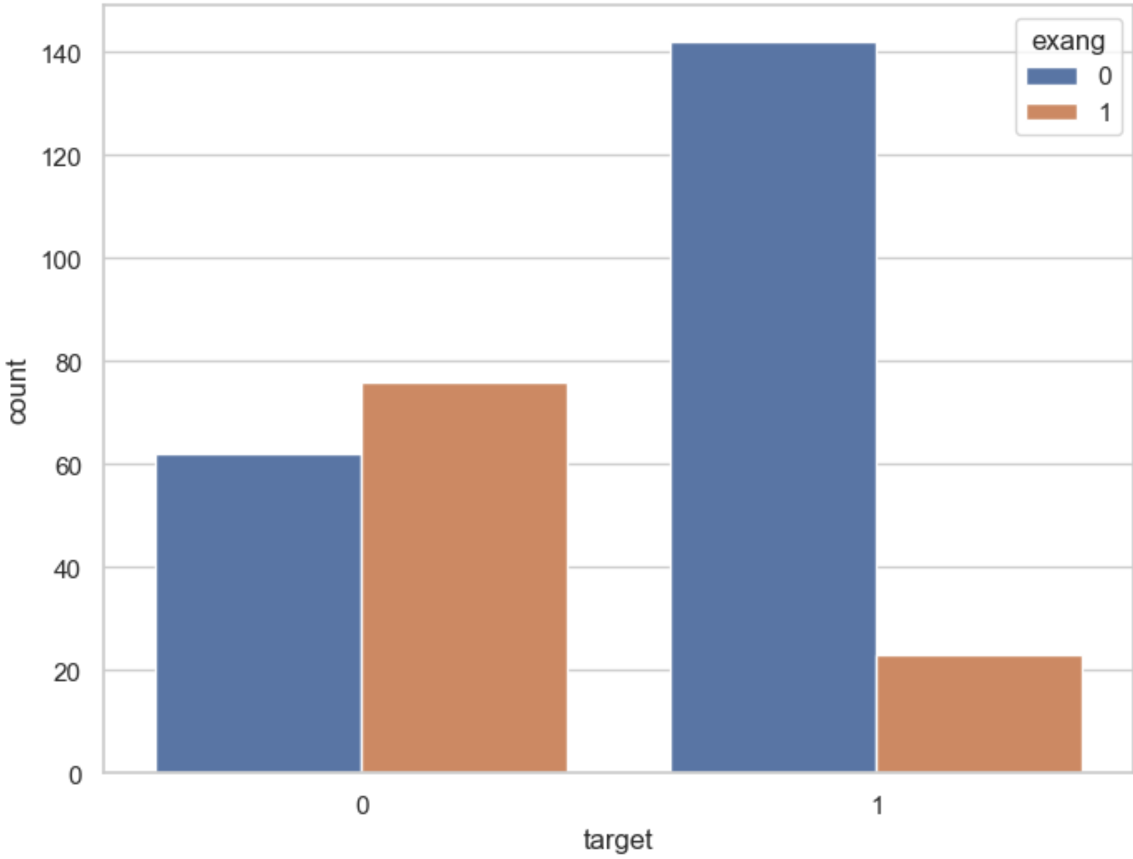
```
ax = sns.countplot(x="target", data=df, facecolor=(0, 0, 0, 0), lin
plt.show()
```



In [52]:
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.countplot(data = df,x = "target",hue = "fbs")
plt.show()
```



In [53]:
```
f ,ax = plt.subplots(figsize = (8,6))
ax = sns.countplot(data =df,x ='target',hue ='exang')
```



In [54]:
```
correlation = df.corr()
```

In [56]:
```
correlation['target'].sort_values(ascending=False)
```

Out[56]:
```
target      1.000000
cp          0.433798
thalach     0.421741
slope       0.345877
restecg     0.137230
fbs        -0.028046
chol       -0.085239
trestbps   -0.144931
age        -0.225439
sex        -0.280937
thal       -0.344029
ca         -0.391724
oldpeak    -0.430696
exang      -0.436757
Name: target, dtype: float64
```

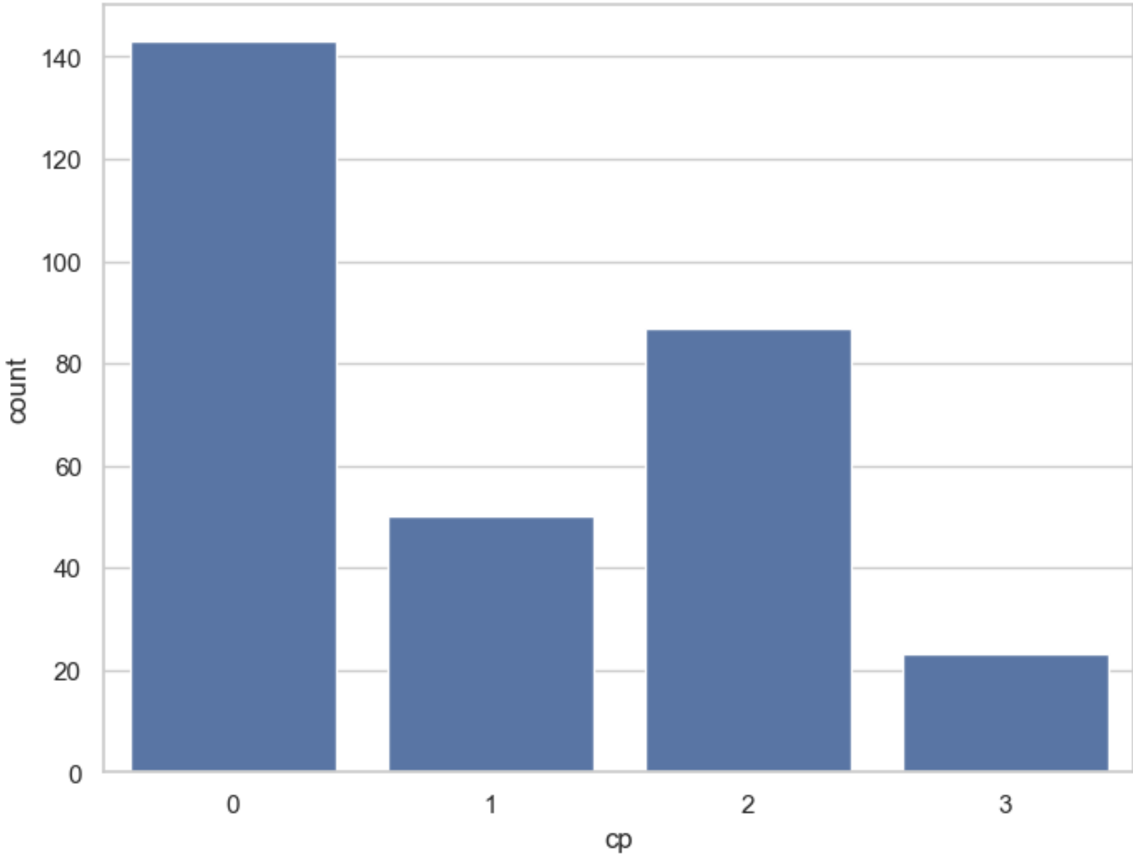
In [57]:
```
df['cp'].nunique()
```

Out[57]:
```
4
```

In [58]:
```
df['cp'].value_counts()
```

Out[58]:
```
cp
0    143
2     87
1     50
3     23
Name: count, dtype: int64
```

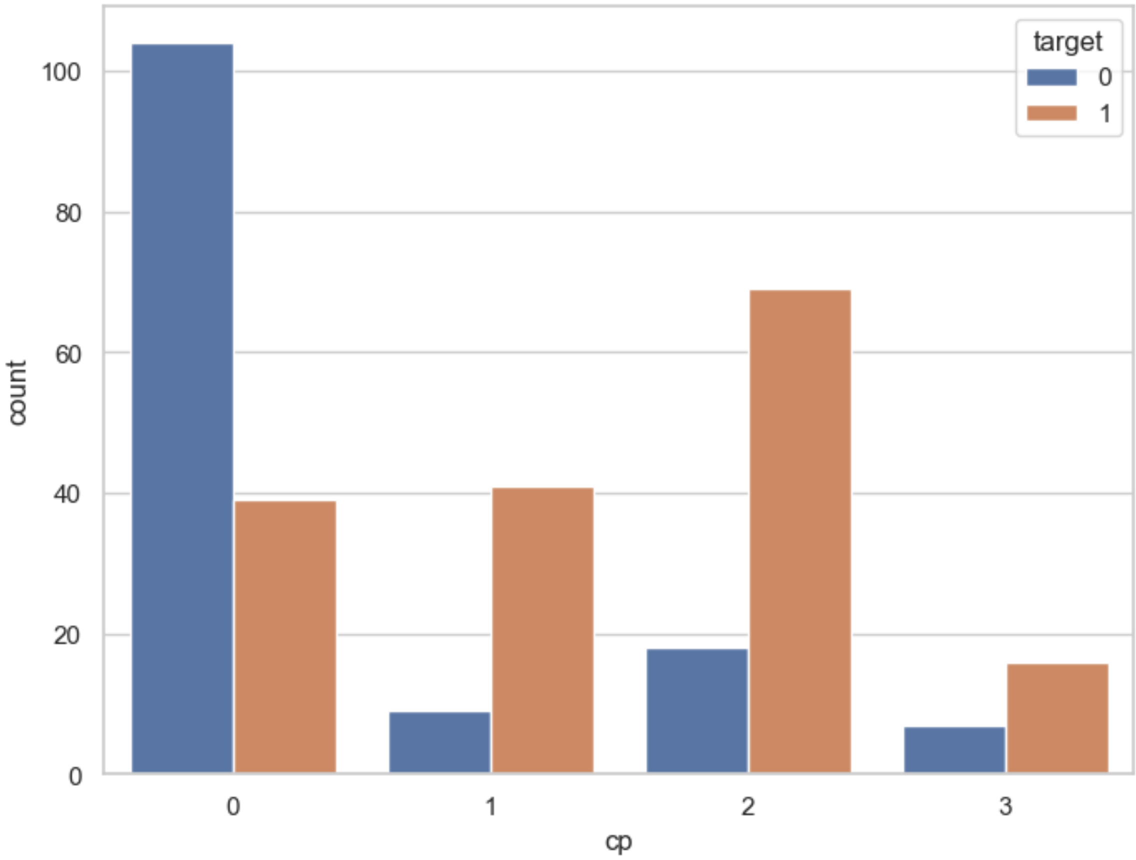
In [60]:
```
f, ax = plt.subplots(figsize=(8,6))
ax = sns.countplot(x ='cp',data =df)
```



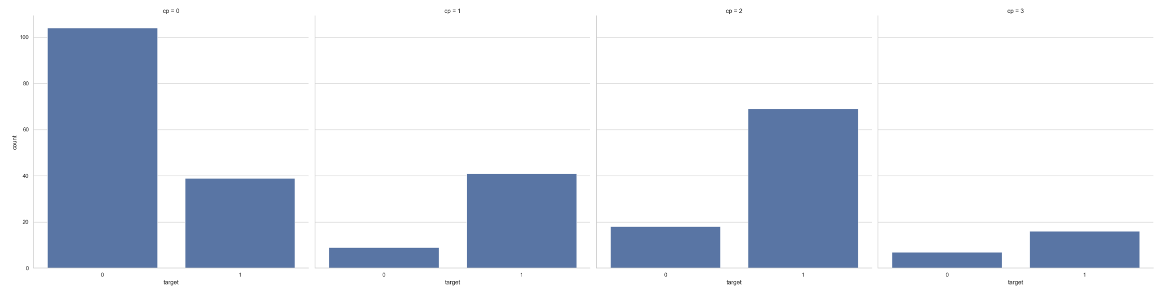
In [61]:
```
df.groupby('cp')['target'].value_counts()
```

Out[61]:
```
cp  target
0   0         104
    1          39
1   1          41
    0           9
2   1          69
    0          18
3   1          16
    0           7
Name: count, dtype: int64
```

In [62]:
```
f, ax =plt.subplots(figsize = (8,6))
ax = sns.countplot(data =df,x ='cp',hue='target')
```



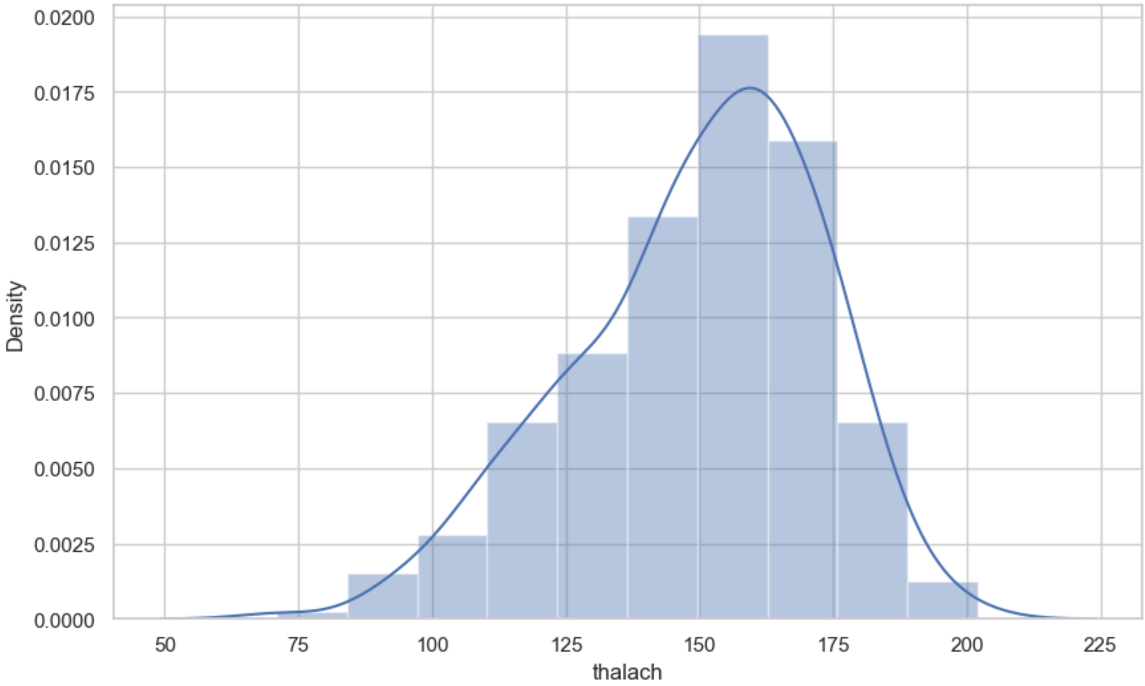
In [63]:
```
ax = sns.catplot(x="target", col="cp", data=df, kind="count", heigh
```



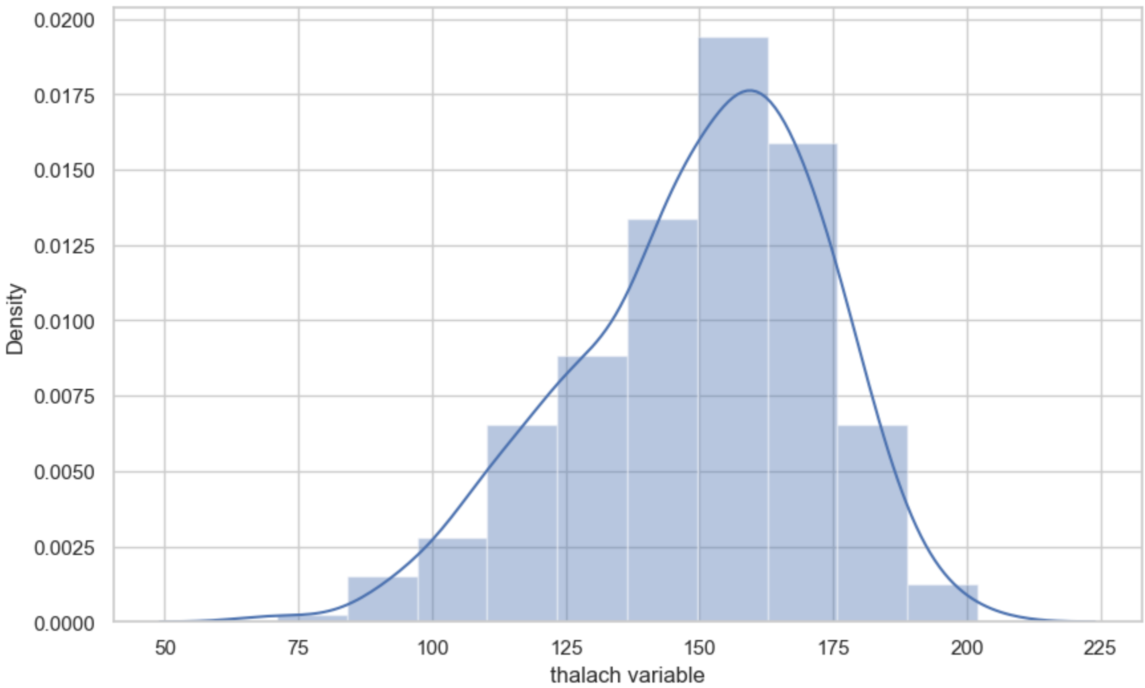
In [64]:
```
df['thalach'].nunique()
```

Out[64]: 91

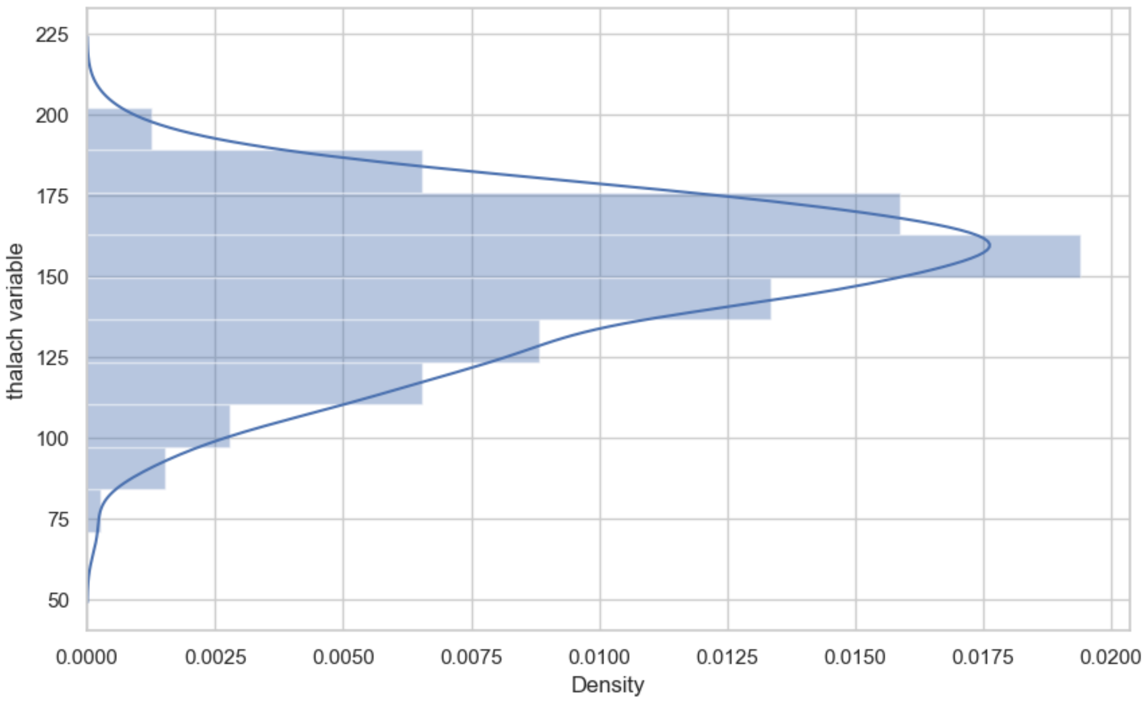
In [66]:
```
f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
ax = sns.distplot(x, bins=10)
plt.show()
```



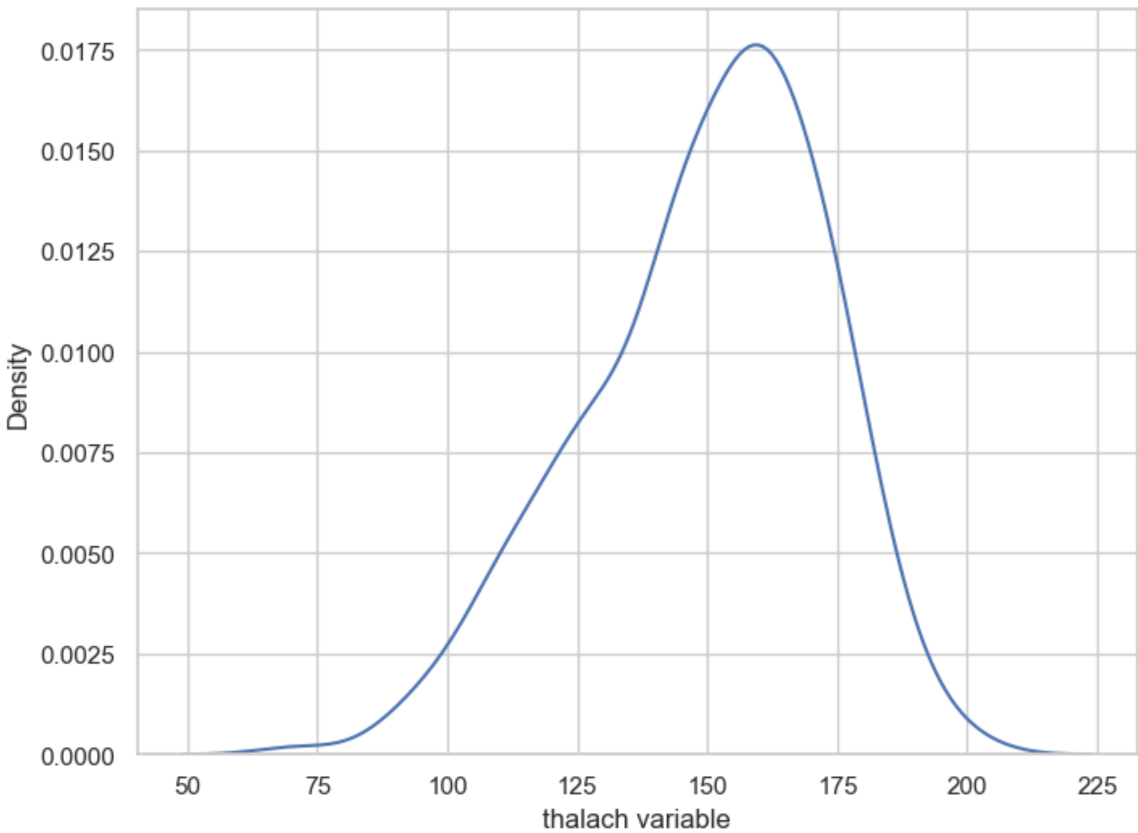
In [67]:
```
f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x,name='thalach variable')
ax =sns.distplot(x,bins=10)
```



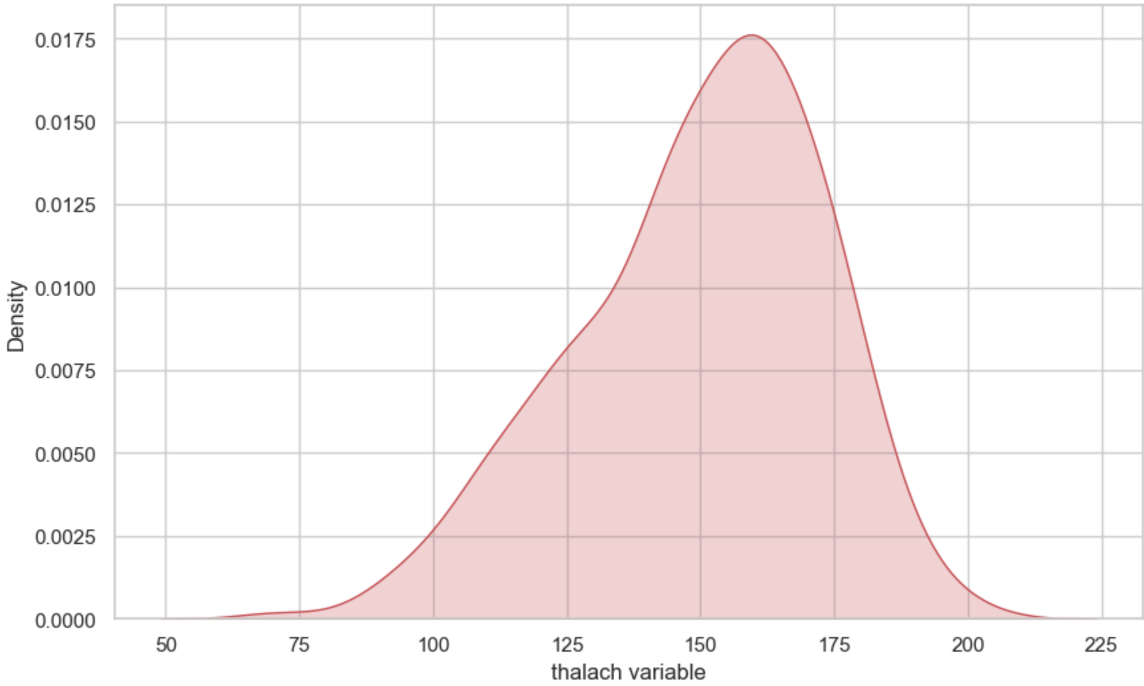
In [68]:
```
f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
ax = sns.distplot(x, bins=10,vertical=True)
plt.show()
```



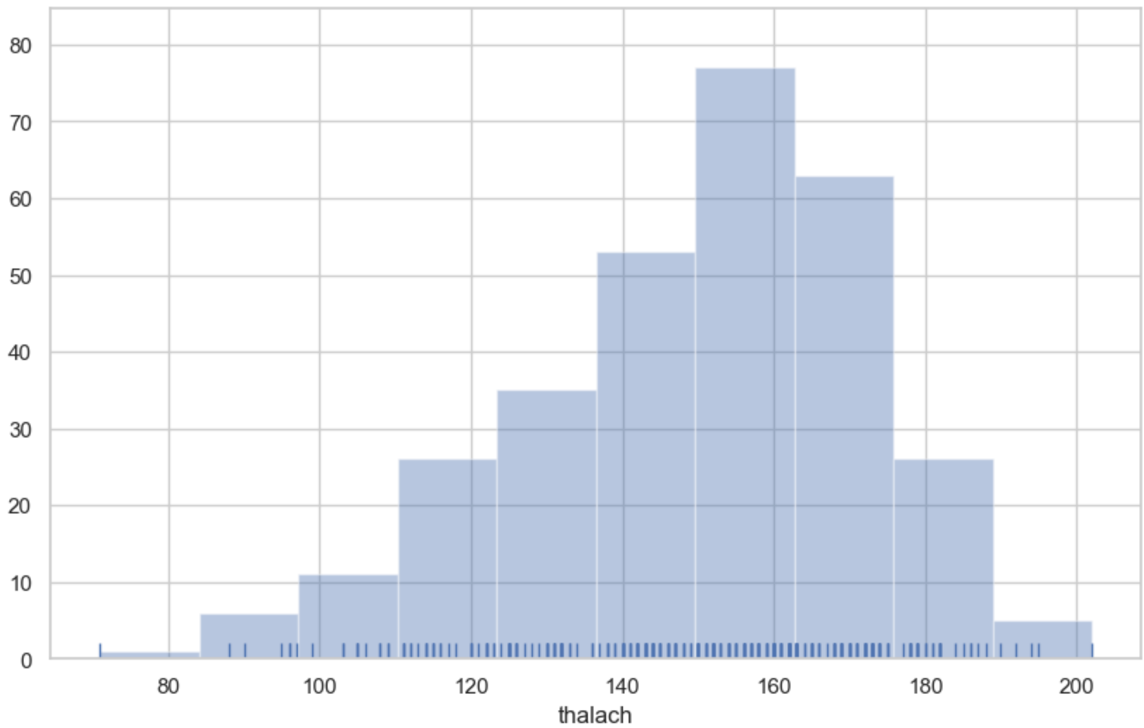
In [69]:
```
f,ax = plt.subplots(figsize=(8,6))
x = df['thalach']
x = pd.Series(x,name='thalach variable')
ax = sns.kdeplot(x)
```



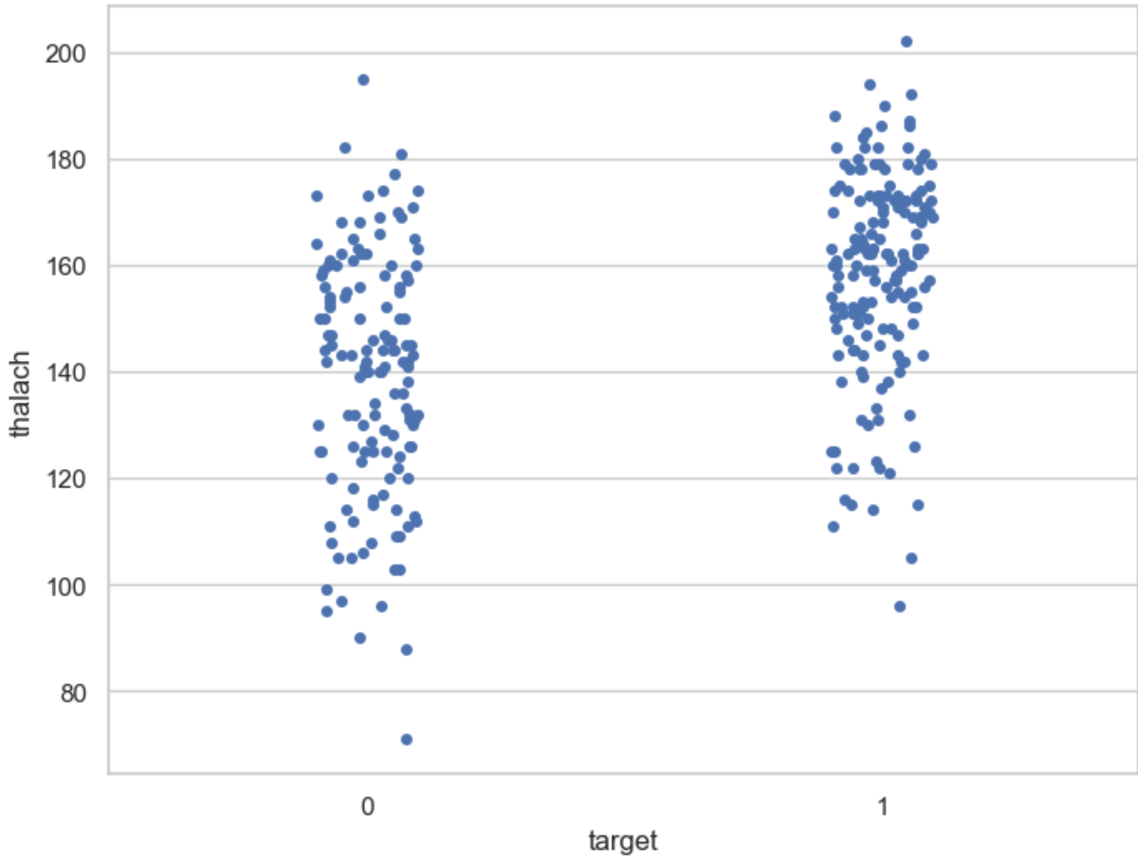
In [70]:
```
f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
x = pd.Series(x, name="thalach variable")
ax = sns.kdeplot(x,shade = True,color='r')
```



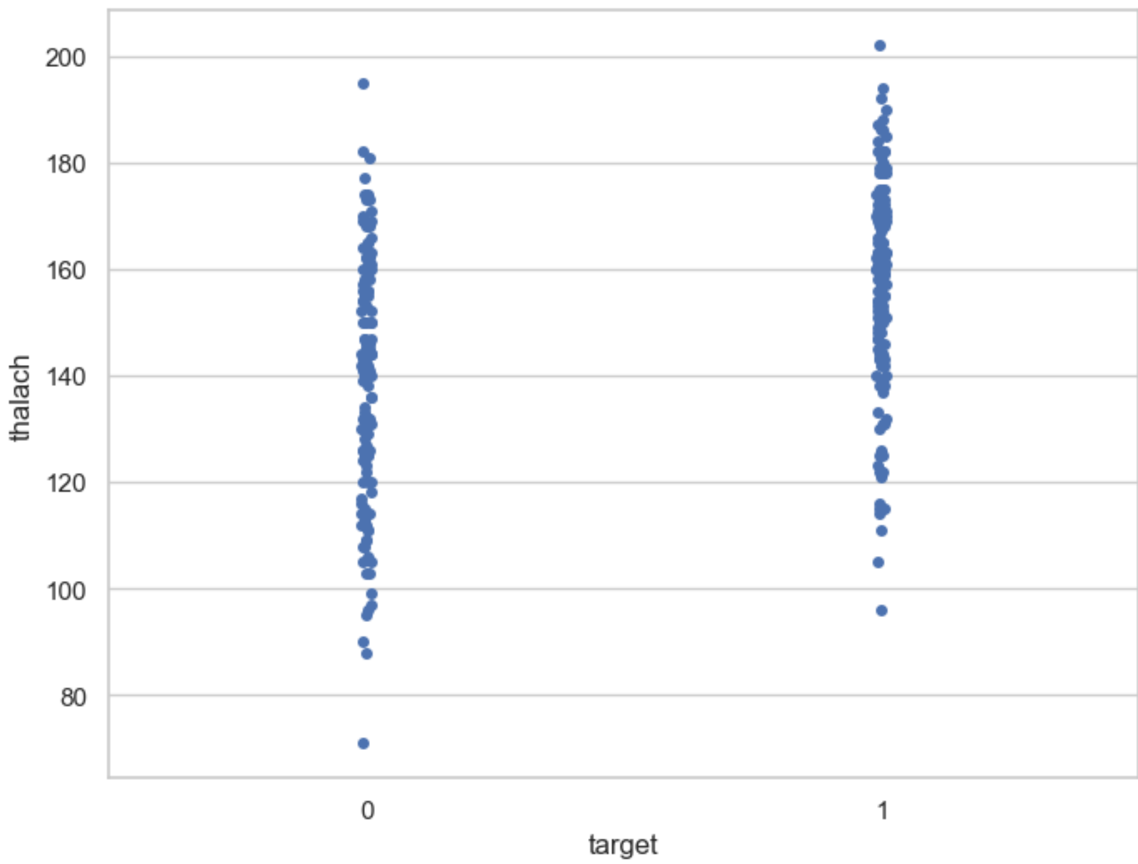
In [71]:
```
f, ax = plt.subplots(figsize=(10,6))
x = df['thalach']
ax = sns.distplot(x, kde=False, rug=True, bins=10)
plt.show()
```



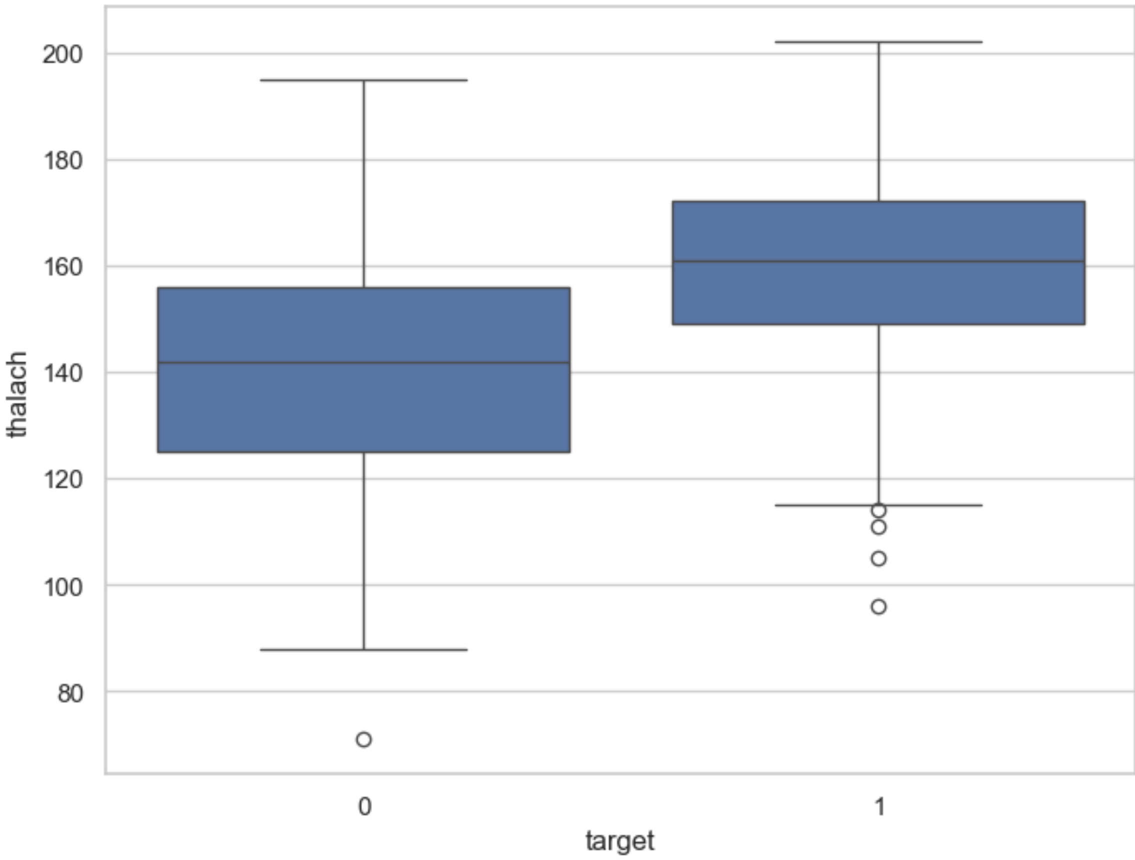
In [72]:
```
f,ax = plt.subplots(figsize=(8,6))
ax = sns.stripplot(data =df,x ='target',y ='thalach')
```



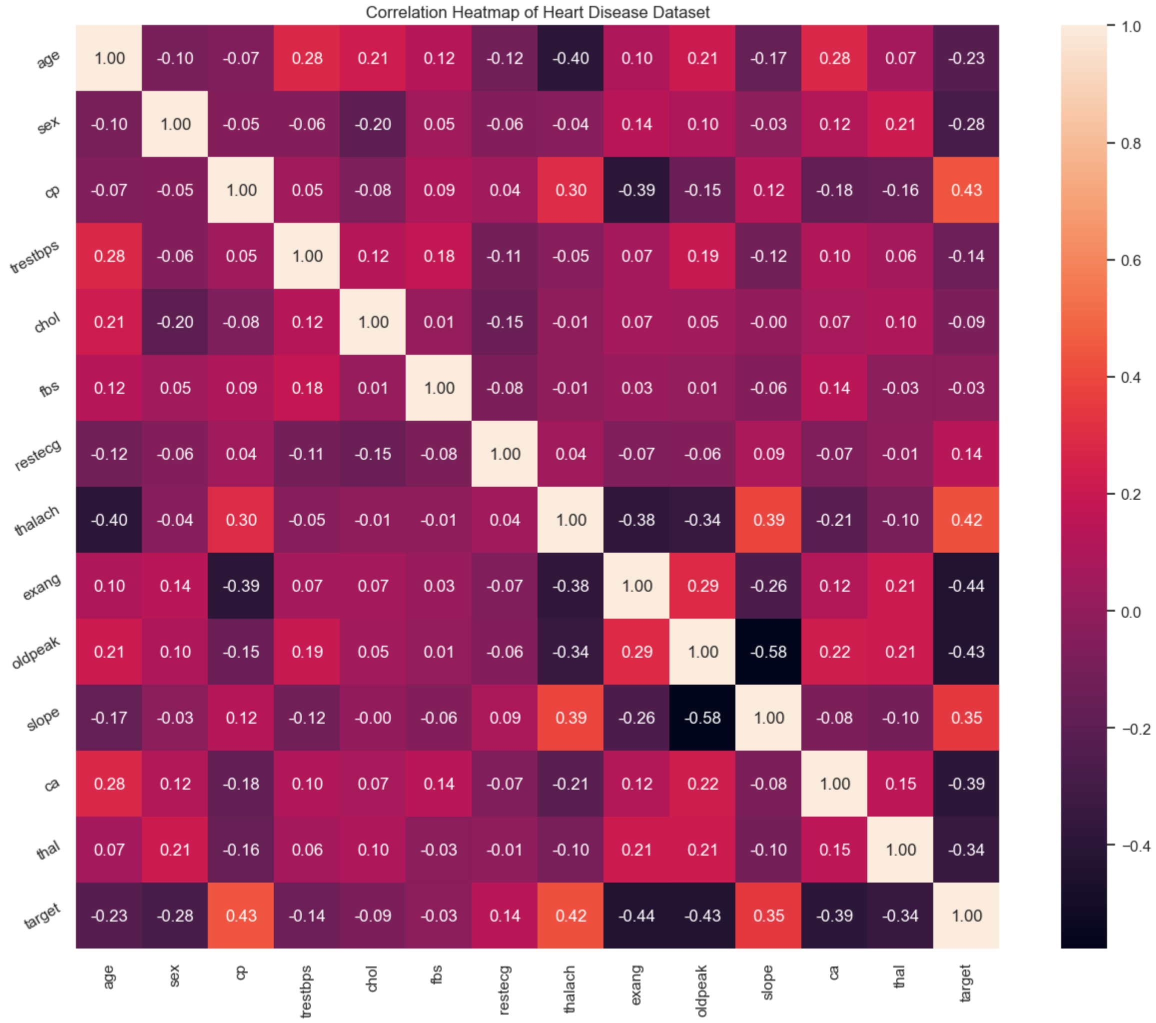
In [73]:
```
f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="target", y="thalach", data=df, jitter = 0.01)
plt.show()
```



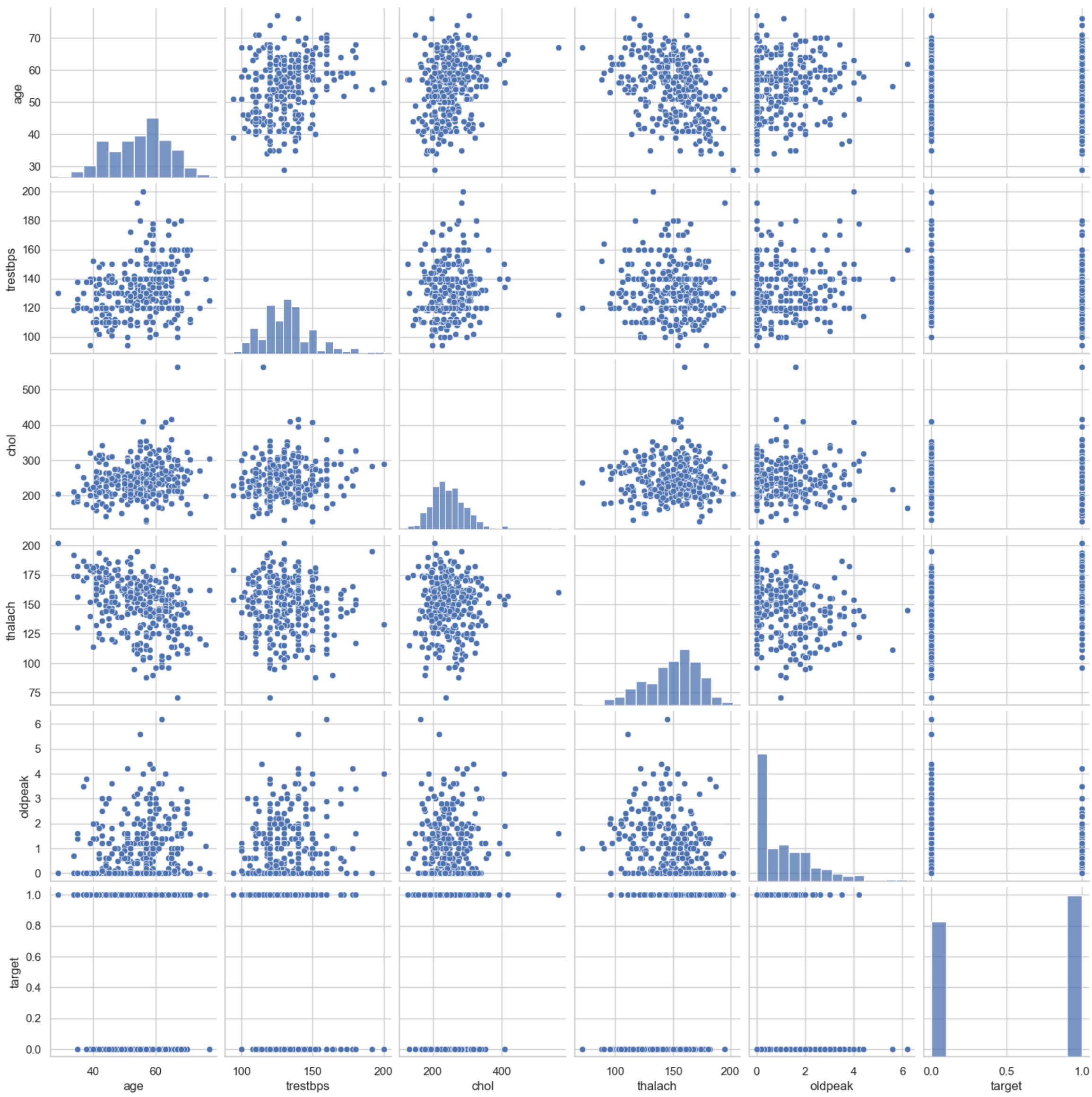
In [75]:
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.boxplot(data =df,x ='target',y ='thalach')
```



In [76]:
```
plt.figure(figsize=(16,12))
plt.title('Correlation Heatmap of Heart Disease Dataset')
a = sns.heatmap(correlation, square=True, annot=True, fmt='.2f', lin
a.set_xticklabels(a.get_xticklabels(), rotation=90)
a.set_yticklabels(a.get_yticklabels(), rotation=30)
plt.show()
```



In [77]:
```
num_var = ['age', 'trestbps', 'chol', 'thalach', 'oldpeak', 'target
sns.pairplot(df[num_var], kind='scatter', diag_kind='hist')
plt.show()
```



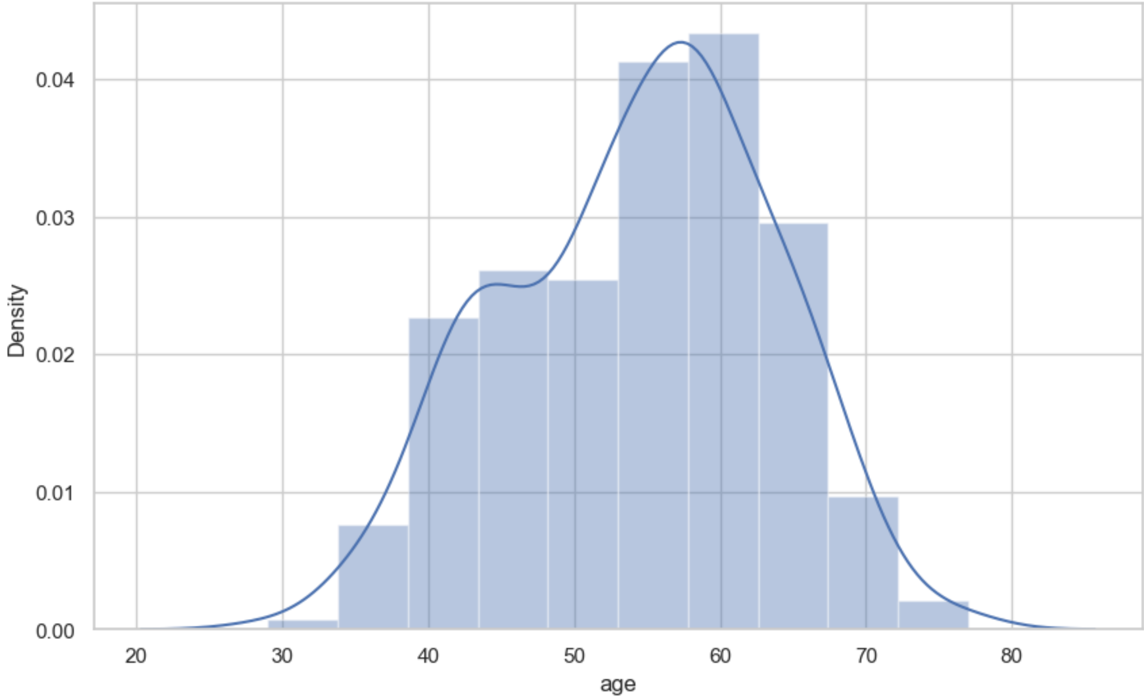
In [78]:
```
df['age'].nunique()
```

Out[78]: 41

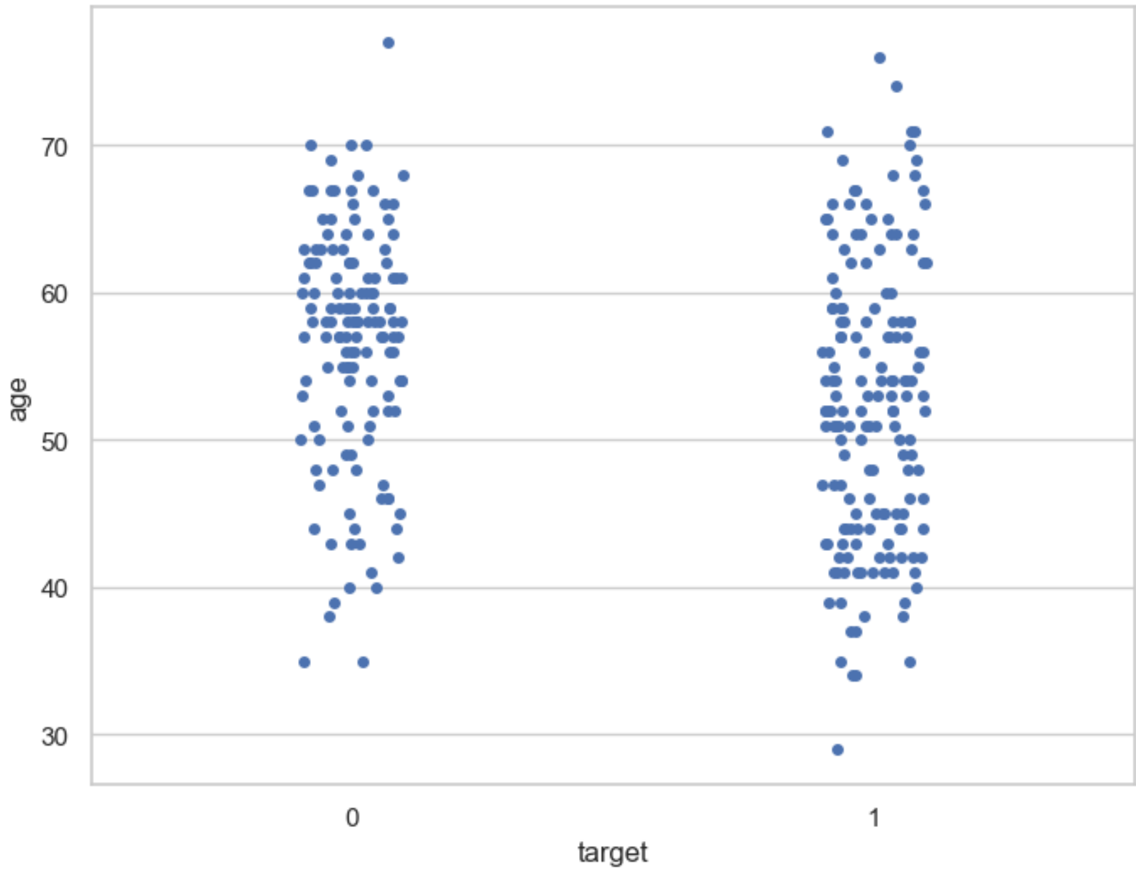
In [79]:
```
df.age.describe()
```

Out[79]:
```
count    303.000000
mean      54.366337
std        9.082101
min       29.000000
25%       47.500000
50%       55.000000
75%       61.000000
max       77.000000
Name: age, dtype: float64
```

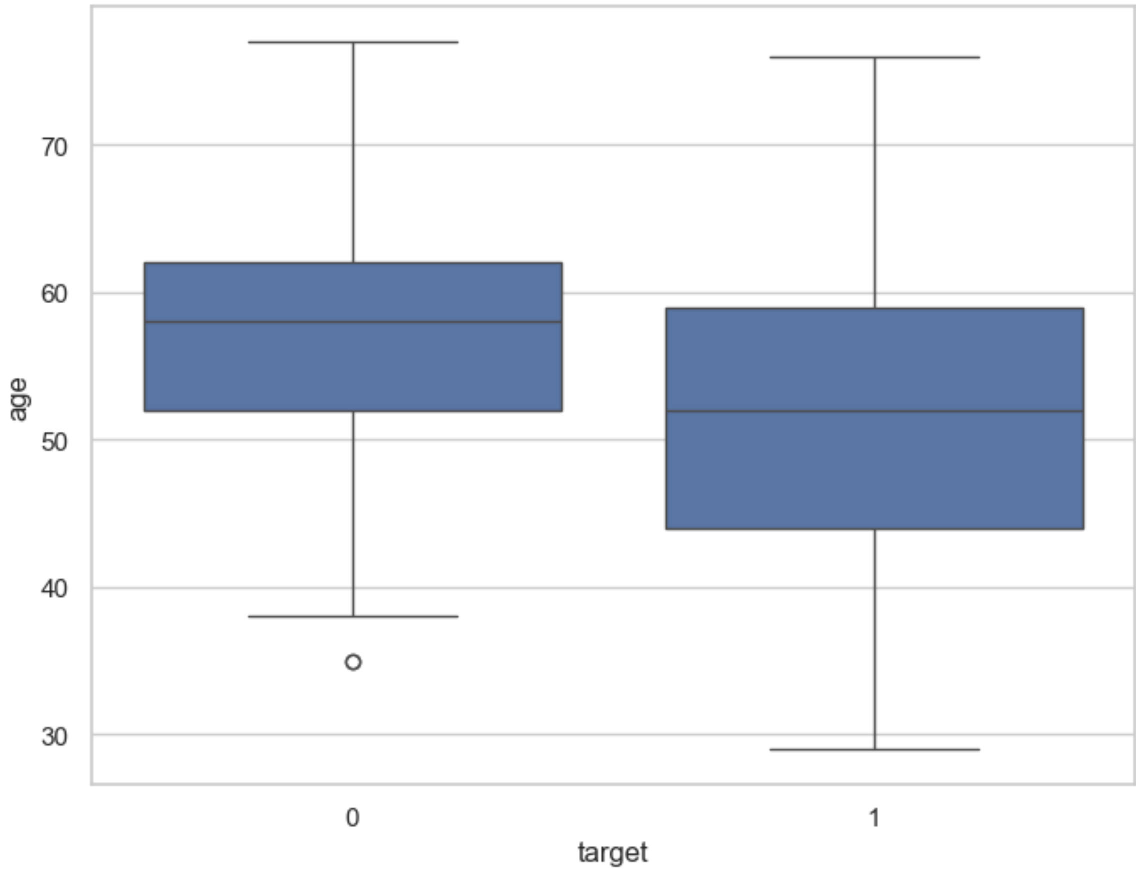
In [81]:
```
f, ax = plt.subplots(figsize=(10,6))
x = df['age']
ax = sns.distplot(x, bins=10)
plt.show()
```



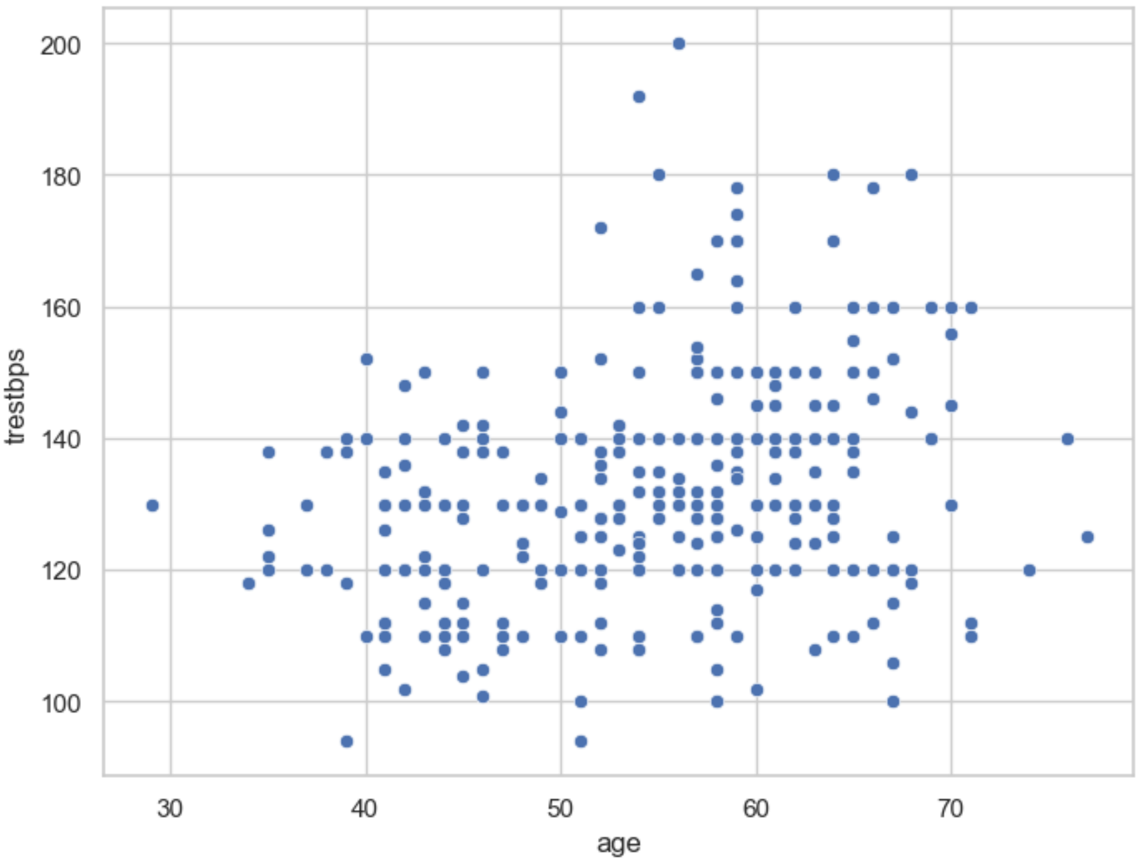
In [82]:
```
f, ax = plt.subplots(figsize=(8, 6))
sns.stripplot(x="target", y="age", data=df)
plt.show()
```



In [83]:
```
f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x="target", y="age", data=df)
plt.show()
```

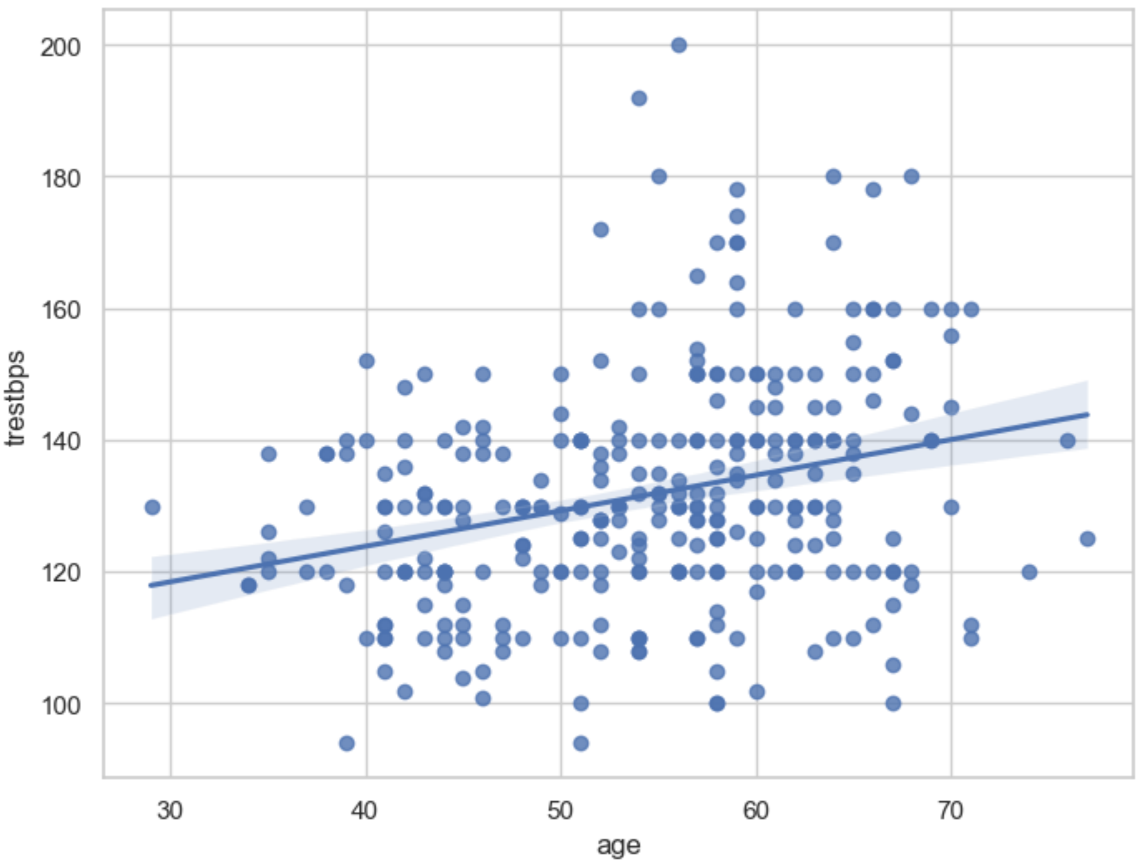


In [84]:
```
f, ax = plt.subplots(figsize=(8, 6))
ax = sns.scatterplot(x="age", y="trestbps", data=df)
plt.show()
```



In [85]:
```
f, ax = plt.subplots(figsize=(8, 6))
```

```
ax = sns.regplot(x="age", y="trestbps", data=df)
plt.show()
```



In [86]:
```
df.isnull().sum()
```

Out[86]:
```
age         0
sex         0
cp          0
trestbps    0
chol        0
fbs         0
restecg     0
thalach     0
exang       0
oldpeak     0
slope       0
ca          0
thal        0
target      0
dtype: int64
```

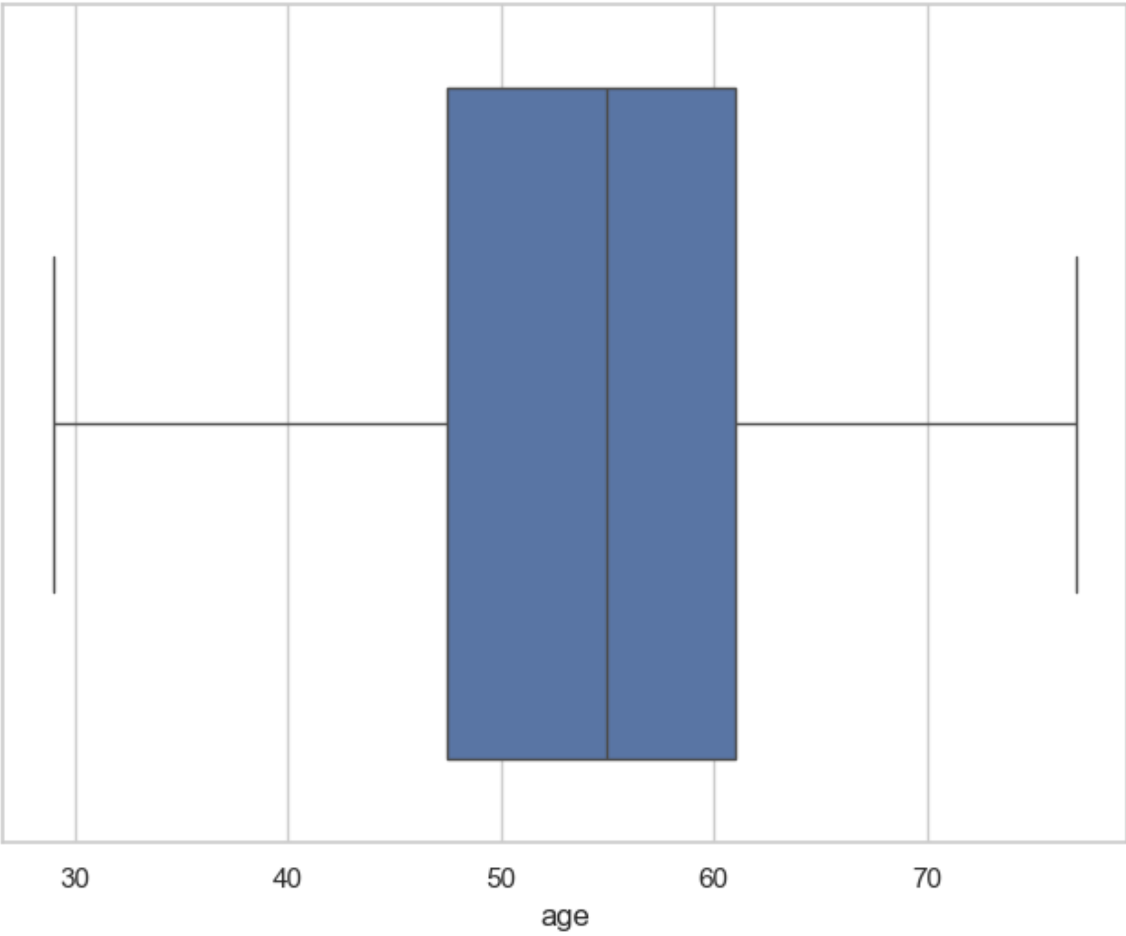
In [87]:
```
assert pd.notnull(df).all().all()
```

In [88]:
```
assert (df >= 0).all().all()
```

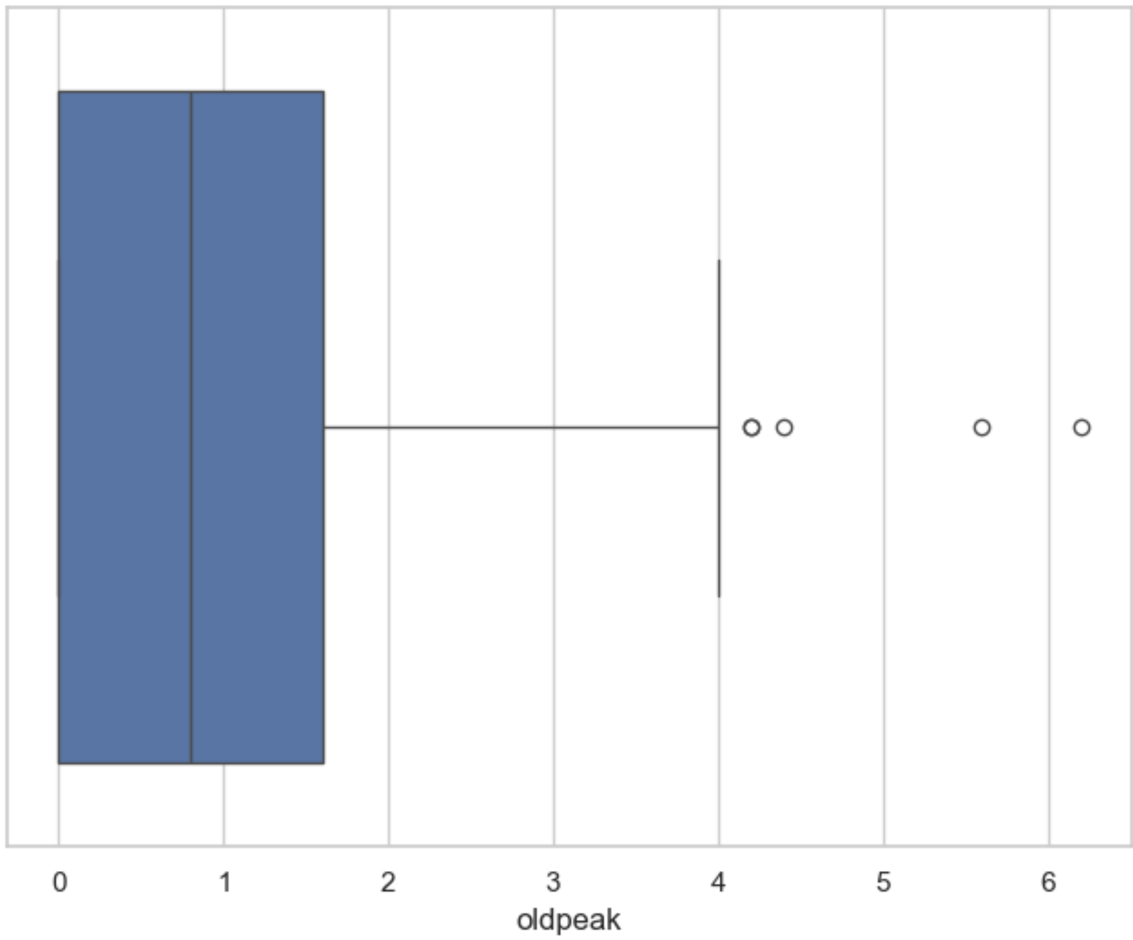
In [89]:
```
df['age'].describe()
```

Out[89]:
```
count    303.000000
mean      54.366337
std        9.082101
min       29.000000
25%       47.500000
50%       55.000000
75%       61.000000
max       77.000000
Name: age, dtype: float64
```

In [90]:
```
f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x=df["age"])
plt.show()
```



In [91]:
```
f, ax = plt.subplots(figsize=(8, 6))
sns.boxplot(x=df["oldpeak"])
plt.show()
```




```
In [ ]:
```